This is what we call a shift from a monolithic codebase to a microservices approach, which is a complex topic worthy of an entire book. The core principle, though, is that both monolithic and microservices architectures serve their purposes depending on the current requirements and context.

11.3 Additional thoughts on AI and the future of web development

As you've seen throughout this book, the landscape of AI-powered web development is evolving at an extremely fast pace. I urge you to continuously invest time and effort in learning and experimenting with these technologies to keep up with the changing tools, libraries, and best practices. The libraries and SDKs we use here are under constant development, and by the time you reach this section, the example codebases may require updates to dependencies or APIs.

What remains constant, however, are the fundamental concepts of web development and the importance of understanding how to use AI effectively. AI is becoming deeply integrated into the core of web technologies, enabling smarter, more personalized, and adaptive user experiences. Trends for 2026 focus on autonomous AI agents performing cross-platform tasks, edge-based inference for near-instant privacy-preserving intelligence, and multimodal optimization as the new standard for accessibility. The future of web development is a transition from "building for the user" to "building environments where AI works for the user." Success belongs to the developers who master these autonomous patterns while maintaining the core pillars of security and performance.

For a final practical example of this forward-thinking approach, the final chapter delves into a real-world integration that prepares your applications for the future of AI.

Summary

- A RAG web application enables users to create, organize, and interact with multiple knowledge bases, each containing uploaded documents such as PDFs and DOCX files.
- The chat interfaces allow users to ask questions and receive answers grounded in the content of their knowledge bases, using LangChain.js retrievers and the Vercel AI SDK for conversational AI.
- Key architectural decisions include considering shared versus dedicated vector stores for user data, implementing secure document handling, and planning for future scalability and modularization.
- The project highlights the importance of designing for security, maintainability, and extensibility. To improve on this basic functionality, consider adding API rate limiting, malware scanning, background processing, and adhering to formal API specifications.